MANIPAL UNIVERSITY JAIPUR

Department of Information Technology

Assignment-1

Program: B Tech Information Technology, VI Semester

Course: INT3201- Computer Networks

Topic: Physical Layer and Data Link Layer (TCP/IP Model)

Q1. Nyquist + Shannon Comparative Analysis

A communication channel has bandwidth 25 kHz and SNR of 30 dB.

1. Calculate maximum data rate using **Nyquist theorem** assuming 16 signal levels.
2. Calculate channel capacity using **Shannon's theorem**.
3. Which value limits the system? Justify logically.

Q2. Multi-Segment End-to-End Delay

A 8 MB file is transmitted over a 3-hop path with the following properties:

Link	Bandwidth	Length
1	50 Mbps	1000 km
2	20 Mbps	2000 km
3	100 Mbps	500 km

Propagation speed = (2.5×10^8) m/s and Packet size = 1500 Bytes. Calculate:

1. Transmission delay per link
2. Propagation delay per link
3. Total end-to-end delay assuming store-and-forward

Q3. Bandwidth-Delay Product + Window Size

A satellite link has:

- Bandwidth = 40 Mbps
- RTT = 500 ms
- Frame size = 2000 Bytes

Calculate:

1. Bandwidth-delay product
2. Minimum window size (in frames) for 100% utilization
3. Buffer size required at sender

Q4. Error Probability and Frame Success

Bit error rate (BER) of a channel is (10^{-6}).
Frame size = 2000 Bytes.

1. Probability that a frame is received without error
2. Probability that at least one bit error occurs
3. If using Stop-and-Wait ARQ, estimate average transmissions per successful frame.

Q5. CRC Division (Higher Order Polynomial)

Given data: 1101011011110001
Generator polynomial: ($G(x) = x^4 + x^3 + x + 1$)

1. Find CRC remainder
2. Give transmitted codeword
3. Show receiver verification process

Q6. Go-Back-N Throughput

Given:

- Data rate = 5 Mbps
- Frame size = 1000 Bytes
- Propagation delay = 50 ms
- Window size = 7

1. Calculate transmission time
2. Calculate utilization
3. Calculate effective throughput

Q7. Selective Repeat vs Go-Back-N Performance

Using data from Q8, assume BER = (10^{-5}).

1. Compute probability of frame error
2. Estimate retransmissions in Go-Back-N
3. Estimate retransmissions in Selective Repeat
4. Compare efficiency numerically

Q8. Signal Attenuation and Power

A signal with initial power 10 mW passes through 3 sections of cable with attenuation:

- 3 dB
- 10 dB
- 7 dB

1. Calculate total attenuation
2. Find final received power
3. If minimum detectable power is 0.1 µW, determine if communication is possible

Q9. Framing Techniques

Compare:

- Byte stuffing
- Bit stuffing

Explain advantages and failure cases.

Q10. CRC vs Checksum Mathematical Strength

Why is CRC mathematically stronger than checksum? Explain in terms of polynomial arithmetic.

Q11. Sliding Window Sequence Numbering

If window size = 7:

1. What is minimum sequence number field size for Go-Back-N?
2. What is minimum sequence number field size for Selective Repeat?
3. Justify mathematically.